

Conversions

Convert the following numbers to their representatives in base 2, 16, & 10.

$$0110\ 0100_2$$

$$= \boxed{64}_{16}$$

binary is hex

$$= 6 \times 16 + 4 \times 1$$

$$= \boxed{100}_{10}$$

$$47_{16}$$

$$= \boxed{0100\ 0111}_2$$

binary is hex

$$= 4 \times 16 + 7 \times 1$$

$$= \boxed{71}_{10}$$

$$47_{10}$$

$$47 \div 16 = \boxed{2} R15$$

$$15 \div 1 = \boxed{15} R0$$

$$= \boxed{2F}_{16}$$

$$= \boxed{0010\ 1111}_2$$

binary is hex

$$1010\ 1010\ 1010_2$$

$$= \boxed{AAA}_{16}$$

binary is hex

$$= 2560 + 160 + 10$$

$$= \boxed{2730}_{10}$$

$$7E3_{16}$$

$$= \boxed{0111\ 1110\ 0011}_2$$

binary is hex

$$= 7 \times 256 + 14 \times 16 + 3 \times 1$$

$$= 1792 + 224 + 3$$

$$= \boxed{2019}_{10}$$

$$999_{10}$$

$$999 \div 256 = \boxed{3} R231$$

$$231 \div 16 = \boxed{14} R7$$

$$7 \div 1 = \boxed{7} R0$$

$$= \boxed{3E7}_{16}$$

$$= \boxed{0011\ 1110\ 0111}_2$$

$$1800_{10}$$

$$1800 \div 256 = \boxed{7} R8$$

$$8 \div 16 = \boxed{0} R8$$

$$8 \div 1 = \boxed{8} R0$$

$$= \boxed{708}_{16}$$

$$= \boxed{0111\ 0000\ 1000}_2$$

binary is hex

$$1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111_2$$

$$= 2^{32} - 1 = \boxed{4,294,967,295}$$

Bonus

Calculate the date that a fixed-point, 32-bit, 2's-complement number of seconds will not be enough to count the seconds from midnight 1970/01/01. (When is 2^{31} seconds from midnight 1970/01/01?)

$$2^{31} \div 60 \div 60 \div 24 \div 365.25 + 1970 \approx \boxed{2038.05}$$

$$\text{Since... } 0.05 \times 365.25 \approx 18$$

The fixed-point, 32-bit, 2's-complement number of seconds will overflow on 2038/01/19 (the so-called *Year 2038 Problem*).
